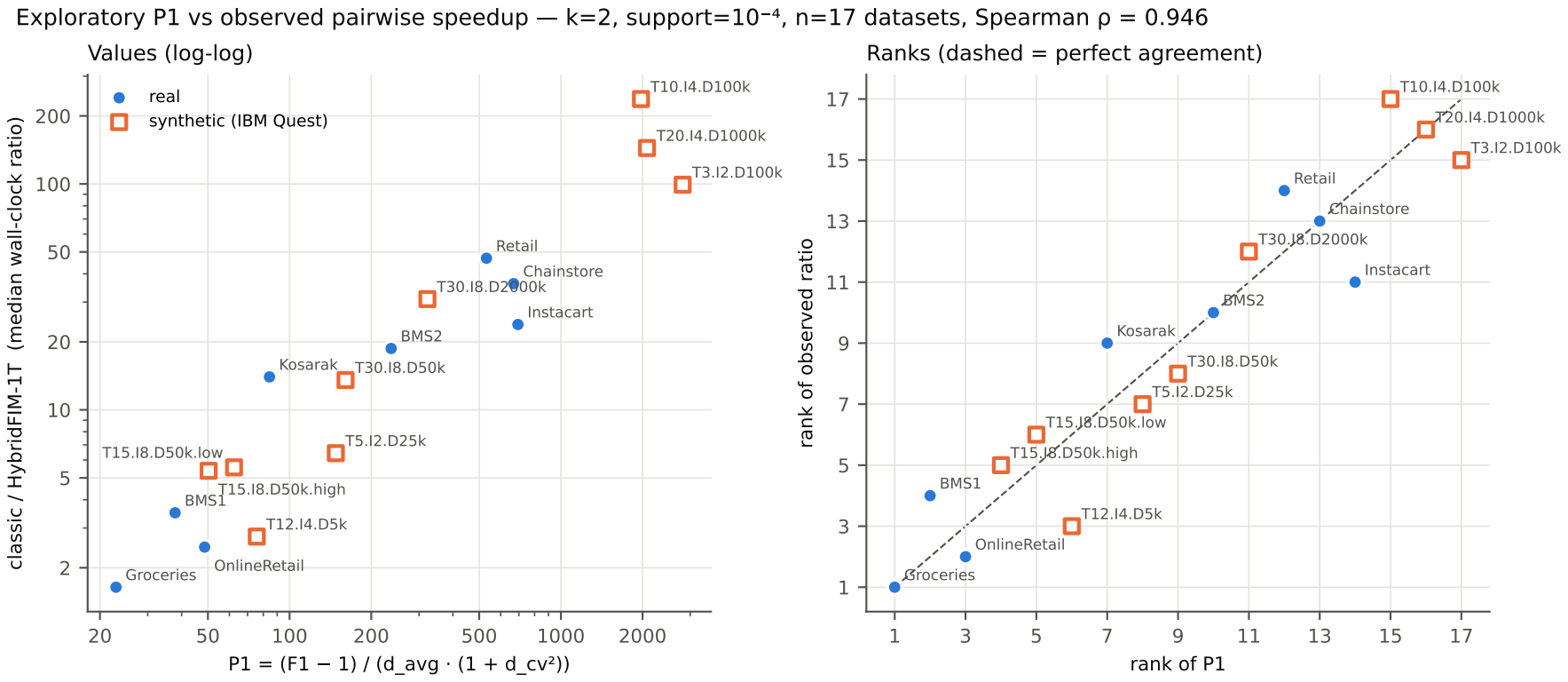
Exploratory P1 vs observed pairwise speedup — k=2, support=10⁻⁴, n=17 datasets, Spearman ρ = 0.946
Values (log-log)
real
synthetic (IBM Quest)
T10.I4.D100k
T20.I4.D1000k
T3.I2.D100k
Retail
Chainstore
T30.I8.D2000k
Instacart
BMS2
Kosarak
T30.I8.D50k
T5.I2.D25k
T15.I8.D50k.low
T15.I8.D50k.high
BMS1
T12.I4.D5k
OnlineRetail
Groceries
classic / HybridFIM-1T (median wall-clock ratio)
P1 = (F1 − 1) / (d_avg · (1 + d_cv²))
Ranks (dashed = perfect agreement)
rank of observed ratio
rank of P1